Berkay **Barlas**

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Technical Skills

AI & Machine Learning LLMs, Multi-modal Models (VLM), On-device AI, CoreML, Embeddings, Semantic Search, LLM Evaluation, Synthetic Data, Py
Programming Languages Swift, Objective-C, C/C++, Python, TypeScript, Go, WebAssembly
Systems & Infrastructure UIKit, SwiftUI, AppKit, React, Node.js, Docker, Kubernetes, AWS

Work Experience

Apple Cupertino, CA
Software Engineer – Apple Intelligence & Safari Intelligence Sept 2025 - Present
- Built an LLM evaluation framework for Safari's first intelligence features, designing evaluation metrics for model quality across **24 locales** with synthetic data generation and prompt evaluation pipelines.
- Reduced evaluation costs by **90%** and cut pipeline runtimes by **67%** through batching, caching, parallelization, and inference optimization.
- Resolved model cold-start and latency issues for on-device LLM inference; decreased power consumption of AI-powered Tab Clustering.
- Designed an on-device scheduler for prioritization and execution of LLM and multi-modal tasks on iOS.

Software Engineer @Safari (iOS) **2+ Years** | July 2023 - Sept 2025
- Developed and optimized Safari browser components on iOS using UIKit, Objective-C, and Swift, serving **1B+ users** worldwide.
- Built an on-device semantic clustering pipeline using foundation embedding models and custom ranking algorithms, enabling automatic tab organization with privacy-preserving on-device inference.

Vagon.io Remote
Founding Frontend Lead **2 Years** | Mar 2020 - Jan 2022
- Led a front-end team building a real-time remote desktop streaming app; optimized WebAssembly and WebSocket pipelines, improving performance by **30%**.
- Reduced monthly fraud by **80%** using anomaly detection with rule-based scoring and statistical thresholds.

Google San Francisco, CA
Software Engineering Intern **3 Months** | Aug 2022 - Nov 2022
- Built a latency-tracking tool for Google Cloud products, aggregating **billions of log entries** via SQL to identify performance bottlenecks and regressions.
- Implemented a TypeScript dashboard with Go back-end for real-time latency visualization across cloud services.

Apple Cupertino, CA
Software Engineering Intern **3 Months** | May 2022 - Aug 2022
- Engineered features for a major Safari release on Safari and WebKit teams, adopted by **20%** of global users; resolved macOS rendering bugs using Objective-C.

Zeplin.io (Y Combinator 2015 Batch) Istanbul, Turkey
Software Engineer **1+ Year** | Jun 2019 - Aug 2020
- Developed shared React and Electron components; led a performance overhaul improving page load speed by **30%**.

Patents

U.S. Patent Application (Pending) 2025 Co-Inventor. System and method for intelligent interface organization using on-device AI in large-scale consumer software platforms (Apple Inc.).

Program Committees

2025 **Judge / Jury Member - Apple Swift Student Challenge (2025)**, Selected by Apple to evaluate global student submissions based on technical quality, creativity, and execution.

Education

Georgia Institute of Technology (GaTech) Atlanta, US
M.Sc Computer Science Aug 2021 - May 2023
GPA 4.0 / 4.0 | **Coursework:** Adv. Machine Learning Systems, Distributed Systems, Big Data Systems, Advanced Internet Systems.

Koc University Istanbul, Turkey
B.S. Computer Eng. Major and Electrical&Electronics Eng. Double Major Oct 2015 - May 2021
Technical University of Munich (TUM), Germany and **Nanyang Technological University (NTU), Singapore Exchange Student**
Received 100% scholarship awarded to **top 300 (0.01%)** among 2 million students.

Projects

Senior Design Project | fundeus.xyz Best EE Project Award, Spring20
Eye Disease Classification System Oct 2020 - Jan 2021
- Trained **ResNet-based CNN** on **10K+ retinal fundus images** to classify cataract, glaucoma, and diabetic retinopathy, achieving **92% accuracy**.
- Deployed end-to-end inference pipeline using **Python, React, AWS** with a clinical-ready web interface for ophthalmologists.